

Matthew Smith

Senior Software Engineer | Federal & Private Sector Experience
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SUMMARY

Senior Software Engineer specializing in building, modernizing and delivering secure, scalable enterprise systems across federal and private sectors. Experienced in Java (Spring Boot), C# (.NET), and Python (Django) and modern TypeScript frameworks, with deep involvement in mission-critical FEMA platforms including PrepToolkit and URT. Proven track record designing RESTful APIs, leading legacy modernization initiatives, delivering containerized services in production environments, and mentoring engineering teams. Strong focus on maintainability, performance, and clean architecture. Actively exploring LLM integration and Retrieval-Augmented Generation (RAG) architectures.

CORE EXPERTISE

Backend: Java (Spring Boot), C# (.NET), Python (Django), REST APIs, MySQL, PostgreSQL, SQL Server, Flyway

Frontend : TypeScript, Svelte, React, Angular, GWT, Bootstrap, Tailwind

System Design: Microservices, API-Driven Design, Event-Driven Patterns, Distributed Session Management, Layered Systems

DevOps & Tools: Docker/Podman, Gradle, Maven, AWS (Lambda, S3), Git, Apache Tomcat

Platforms: Liferay 7, Node.js, npm

PROFESSIONAL EXPERIENCE

Senior Software Engineer | 4 Arrows Consulting, Inc. • Somerset, KY (Remote) • 03/2019 – Present

- **Cross-Stack Feature Ownership:** Delivered complex features across backend (Spring Boot/.NET), database, and frontend layers within FEMA preparedness platforms (PrepToolkit, OneResponder, RTLT, IRIS), ensuring cohesive integration between APIs, data models, and user interfaces.
- **System Architecture & Presence:** Spearheaded the implementation of a real-time presence tracking system for the Unified Reporting Tool (URT); independently designed and optimized distributed session logic (polling/timeouts) to ensure data integrity for concurrent federal users.
- **Serverless Portal Ownership:** Replaced legacy GWT portlets with a standalone serverless AWS application (S3/CloudFront hosted Svelte frontend) serving as a unified access portal for FEMA preparedness platforms (PrepToolkit, OneResponder, RTLT, IRIS), organizing the component hierarchy (containers, forms, cards) and structuring the provider layer per application to keep each platform's data context cleanly separated.
- **Development Workflow Improvements:** Leveraged containerized development environments (Docker/Podman) and Flyway-based database migrations to improve consistency across environments and reduce configuration drift. Regularly conduct peer code reviews to reinforce clean architecture and maintainable code practices.
- **Stakeholder Alignment:** Translate complex FEMA business requirements (THIRA/SPR, CPG, NIMS, UAWG) into scalable technical solutions, coordinating closely with project leads to ensure all features align with federal compliance and security standards.
- **Operational Excellence:** Automated enterprise reporting workflows (Docmosis/Apache POI), reducing processing time from hours to minutes and directly improving operational efficiency for national exercise programs.

Tech Stack: Java, Spring Boot, Gradle, GWT, MySQL, Liferay 7, Apache HTTP Server, JavaScript/TypeScript, Svelte, React, Node.js, npm, AWS Lambda, Git, Podman, Docmosis, OpenCSV, Apache POI

Applications Developer | United Parcel Service (UPS) • Louisville, KY • 03/2017 – 03/2019

- **Mission-Critical Reliability:** Engineered backend logic and UI features for real-time transportation visibility systems, ensuring operational uptime in a 24/7 high-throughput production environment.

- **Business Logic Abstraction:** Architected complex decision-making flows by translating high-level business requirements into discrete rules for the Drools inference engine (BRMS), reducing manual intervention and decoupling policy from code.
- **Performance Engineering:** Conducted deep-dive bottleneck analysis and performance tuning (JMeter), identifying and resolving critical latency issues under peak traffic conditions.

Tech Stack: Java, Spring Framework, Maven/Gradle, Drools (BRMS), JMeter, Roadrunner, WebLogic, ActiveMQ, SQL Server, TFVC

Software Developer | Apax Software • Lexington, KY • 03/2016 – 03/2017

- **Technical Consultation:** Partnered directly with stakeholders (Presbyterian Church (USA), University of Kentucky) to translate vague business needs into production-ready technical specifications and RESTful architectures.
- **Modular Architecture:** Designed reusable backend components and modular UI features, significantly accelerating release cycles and ensuring scalable feature expansion across multiple client projects.
- **Security & Governance:** Implemented robust server-side validation, authentication, and role-based access control (RBAC) for sensitive event registration and CMS workflows.

Tech Stack: Python, Django, JavaScript (AngularJS, jQuery), PostgreSQL, Bootstrap, AWS, Git, PHP, Drupal

Programming Analyst | Knowledge Facilitations Group (KFG) • Richmond, KY • 03/2015 – 03/2016

- **Data Pipeline Engineering:** Developed Python-based transformation engines to validate and map disparate industry standards (X12, EDIFACT, XML, JSON), ensuring seamless data exchange across the EDITRACE platform.
- **Operational Observability:** Built real-time monitoring dashboards that provided immediate insight into automated data pipelines, reducing troubleshooting time for critical EDI processes.

Tech Stack: Python, EDI (X12, XML, EDIFACT, Tradacoms, JSON, IDoc), Bootstrap, Git

EDUCATION

B.S. in Computer Science, 2015 | Eastern Kentucky University, Richmond, KY

Member, Upsilon Pi Epsilon Honors Society for the Computing and Information Disciplines